

AI Brand OS — Developer Friendly End-to-End Architecture Plan

This platform is essentially:

“HubSpot + Hootsuite + Jasper + Zapier + AI Brand Brain”

The system has 8 major layers:

- 1. Authentication
- 2. AI Brand Learning Engine
- 3. Content Orchestration Engine
- 4. Publishing & Scheduling
- 5. Analytics Dashboard
- 6. Lead Capture + Forms
- 7. CRM + Integrations
- 8. AI Infrastructure Layer

1. SYSTEM ARCHITECTURE

Recommended Stack

Layer	Tech
Frontend	Next.js + Tailwind + ShadCN
Backend API	FastAPI or Node.js (NestJS preferred)
Database	PostgreSQL
Vector DB	Pinecone / Weaviate
Queue System	Redis + BullMQ
Auth	Clerk/Auth0/Supabase
AI	OpenAI + Claude + Gemini

Social APIs	Unipile
Hosting	Vercel + Railway/AWS
Storage	S3/Cloudflare R2
Analytics	PostHog + GA4
Forms	Internal builder
SEO Content	Placid + LLMs
Workflow Engine	Temporal / BullMQ

2. HIGH LEVEL MICROSERVICES

Frontend App



API Gateway



| Auth Service
| Brand Brain Service
| Content Engine
| Scheduling Engine
| Analytics Engine
| CRM Integration Service
| Form Builder Service
AI Agent Service



Database + Queue + Vector DB

3. DATABASE DESIGN

USERS

users

- id

- name
 - email
 - password_hash
 - company_name
 - plan
 - created_at
-

BRANDS

brands

- id
 - user_id
 - brand_name
 - website
 - industry
 - tone
 - temperature
 - persona
 - guardrails
 - created_at
-

BRAND KNOWLEDGE

brand_knowledge

- id
 - brand_id
 - source_type
 - content
 - embeddings
 - created_at
-

SOCIAL CONNECTIONS

social_accounts

- id
- brand_id
- platform
- access_token

- refresh_token
 - metadata
-

CONTENT

content

- id
 - brand_id
 - platform
 - content_type
 - prompt
 - generated_content
 - status
 - scheduled_time
 - published_url
-

ANALYTICS

analytics

- id
 - brand_id
 - platform
 - impressions
 - engagement
 - clicks
 - leads
 - conversions
 - recorded_at
-

FORMS

forms

- id
- brand_id
- name
- schema_json
- embed_code

4. AI BRAND LEARNING ENGINE

This is the core moat.

FLOW

User Inputs Website + Docs + Brand Info



Crawler Extracts Content



LLM Summarization



Tone Analysis



Persona Generation



Embedding Creation



Store in Vector DB



Brand Brain Ready

AI BRAND LEARNING ALGORITHM

Step 1 — Crawl Website

Input

```
{  
  "website": "https://brand.com"  
}
```

Process

- Crawl:
 - Home
 - About
 - Blogs
 - Services
 - Case Studies

Tools

- Firecrawl
 - Crawl4AI
 - Puppeteer
-

Step 2 — Clean Text

Algorithm:

```
def clean_text(html):  
    remove_navbar()  
    remove_footer()  
    remove_ads()  
    remove_scripts()  
    chunk_text()
```

Step 3 — Extract Brand DNA

LLM Prompt:

Analyze this company and generate:

1. Brand tone
 2. Writing style
 3. Emotional positioning
 4. Audience type
 5. CTA style
 6. Brand personality
 7. Forbidden topics
 8. Messaging style
-

Step 4 — Generate Brand Persona

Output Example:

```
{  
  "tone": "Professional + optimistic",  
  "persona": "Climate-tech visionary",  
  "writing_style": "Data-driven storytelling",  
  "guardrails": [  
    "No political claims",  
    "No exaggerated promises"  
  ]  
}
```

Step 5 — Embeddings

Convert all knowledge into embeddings.

chunk → embed → vector DB

Use:

- text-embedding-3-large
 - BGE embeddings
-

RAG SEARCH ALGORITHM

Whenever content is generated:

User Request



Retrieve Relevant Brand Chunks



Retrieve Past Best Performing Posts



Inject Brand Persona

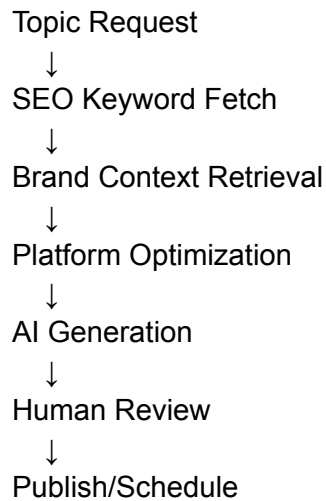


Generate Content

5. CONTENT ORCHESTRATION ENGINE

This is the AI content factory.

CONTENT GENERATION FLOW



BLOG GENERATION ALGORITHM

INPUT

```
{  
  "topic": "Green Hydrogen in India",  
  "platform": "blog",  
  "seo": true  
}
```

PROCESS

Step 1 — SEO Research

APIs:

- Ahrefs
- Semrush
- Google Trends

Fetch:

- Keywords
 - Search volume
 - Competitor blogs
-

Step 2 — SERP Analysis

Extract:

Top ranking headers

Common FAQs

Keyword density

Backlink patterns

Step 3 — Blog Structure Generation

LLM creates:

H1

H2s

FAQs

CTA

Internal links

Meta description

Step 4 — Generate Images

Use:

- Placid
 - Bannerbear
 - Ideogram
-

Step 5 — Humanization Layer

Run:

AI detection reduction
Sentence variation
Brand tone injection

LINKEDIN POST ENGINE

ALGORITHM

Input Topic
↓
Hook Generation
↓
Story Framework
↓
Authority Layer
↓
CTA Layer
↓
Brand Tone Injection

VIRALITY SCORING

Before publishing:

score =
(hook_strength * 0.3) +
(readability * 0.2) +

$(\text{emotional_pull} * 0.2) +$
 $(\text{shareability} * 0.3)$

If score < threshold:
→ regenerate

COMMENT GENERATION ENGINE

FLOW

Fetch Trending Posts



Analyze Context



Generate Smart Comment



Human Approval



Post

6. SOCIAL MEDIA INTEGRATION

Use Unipile

Platforms:

- LinkedIn
 - Instagram
 - X
 - Facebook
-

POST PUBLISHING FLOW

Content Approved
↓
Convert per platform format
↓
Upload media
↓
Publish API
↓
Save Post ID

7. AI SCHEDULING ENGINE

BEST TIME ALGORITHM

Use:

Historical engagement
Audience timezone
Platform trends
Industry benchmarks

ALGORITHM

```
best_time =  
weighted_average(  
    historical_ctr,  
    engagement_rate,  
    platform_peak_hours  
)
```

AUTO CALENDAR ENGINE

Generate Monthly Content Plan



Cluster Topics



Distribute Platforms



Auto-fill Calendar

8. ANALYTICS DASHBOARD

DATA SOURCES

- Google Analytics
 - Social APIs
 - SEO APIs
 - Form submissions
-

METRICS

Views

CTR

Engagement

Leads

CAC

Top performing content

Traffic source

INSIGHT ENGINE

AI analyzes:

What content works

Best platform

Best posting time
Lead conversion drivers

EXAMPLE AI INSIGHT

Posts with founder stories generate 42% more engagement.
Carousel posts perform best on Tuesdays.

9. FORM BUILDER ENGINE

FEATURES

- Drag/drop form builder
 - Conditional logic
 - AI form optimization
 - Embed forms
-

FORM OPTIMIZATION ALGORITHM

Analyze:

- Drop-offs
- Time spent
- Conversion %

Then:

- Reduce fields
 - Reorder questions
 - Improve CTA
-

10. CRM + INTEGRATIONS

CONNECTORS

- HubSpot
 - Salesforce
 - Google Analytics
 - Slack
 - Zapier
 - Webhooks
-

EVENT PIPELINE

Lead Captured
↓
Webhook Trigger
↓
CRM Push
↓
Slack Notification
↓
Email Sequence

11. AI AGENT LAYER

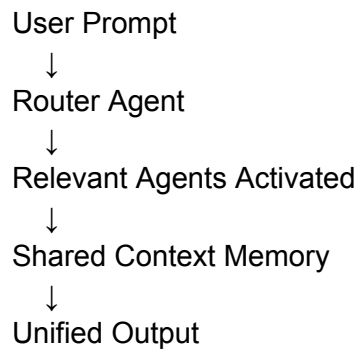
This becomes your future moat.

AGENTS

Agent	Role
Brand Agent	Understands company
SEO Agent	Keyword optimization

Social Agent	Platform content
Analytics Agent	Insights
Growth Agent	Suggests experiments
Comment Agent	Social engagement

MULTI AGENT ORCHESTRATION



12. RECOMMENDED MVP ROADMAP

PHASE 1 (2–4 Weeks)

MVP

Build:

- Auth
 - Brand learning
 - LinkedIn post generation
 - Scheduling
 - Dashboard basics
-

PHASE 2

Add:

- Blog generation
 - Instagram
 - Forms
 - Analytics
-

PHASE 3

Add:

- AI agents
 - Auto-commenting
 - AI insights
 - Backlink engine
-

13. SCALABILITY PLAN

Queue Everything

Never directly process:

- AI generations
- Publishing
- Crawling
- Analytics

Use queues.

CACHE STRATEGY

Cache:

Brand persona

Popular prompts

Analytics snapshots

Embeddings

RATE LIMITING

Required for:

- Social APIs
 - LLM APIs
 - SEO APIs
-

14. SECURITY

MUST HAVE

OAuth encryption

Token rotation

Webhook validation

RBAC

Audit logs

GDPR compliance

15. SUGGESTED FOLDER STRUCTURE

/apps

 /web

 /api

/services

 /brand-engine

/content-engine
/scheduler
/analytics
/integrations

/packages
/ui
/ai-prompts
/shared-types

16. AI PROMPT ARCHITECTURE

SYSTEM PROMPT

You are the AI content brain for {brand_name}.

Brand tone:
{tone}

Audience:
{audience}

Guardrails:
{guardrails}

Writing examples:
{examples}

17. CORE DIFFERENTIATOR

Your biggest moat is NOT content generation.

It is:

Persistent Brand Memory
+
Cross-platform learning

- +
 - Performance feedback loops
 - +
 - AI brand consistency
-

18. IDEAL DEVELOPMENT ORDER

WEEK 1

- Auth
 - Database
 - Brand onboarding
-

WEEK 2

- AI brand learning
 - Embeddings
 - RAG
-

WEEK 3

- LinkedIn generation
 - Scheduling
 - Unipile
-

WEEK 4

- Dashboard
 - Analytics
 - Forms
-

19. FUTURE FEATURES

Advanced AI

- AI video generation
 - AI ad creatives
 - AI voice cloning
 - Autonomous posting
 - Competitor monitoring
 - AI campaign manager
-

20. FINAL RECOMMENDED ARCHITECTURE

Frontend (Next.js)



API Gateway



Agent Orchestrator



| Brand Brain
| Content Engine
| Scheduler
| Analytics
CRM Layer



Postgres + Redis + Vector DB



External APIs + LLMs